

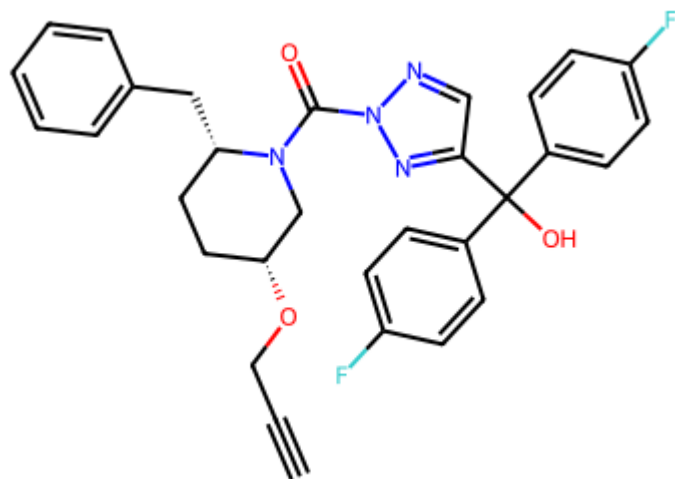


MCS-Grouped Retrosynthesis Report

Predictions grouped by Maximum Common Substructure (MCS) Tanimoto similarity of the largest reactant fragment — distance threshold = 0.50. All structures are aligned to the product via the MCS core.

Product 1

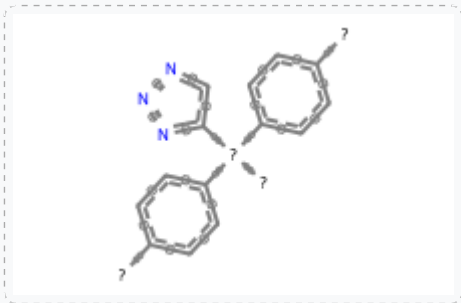
C#CCO[C@@H]1CC[C@H](Cc2ccccc2)N(C(=O)n2ncc(C(=O)(c3ccc(F)cc3)c3cc(F)cc3)n2)C1



10 predictions → 2 MCS match groups

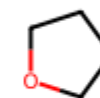
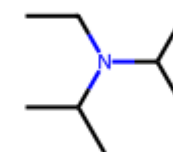
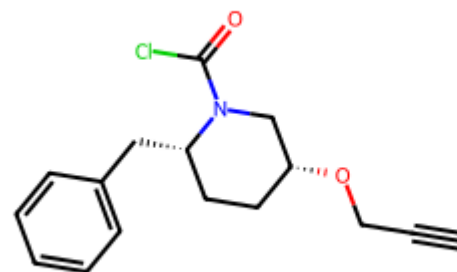
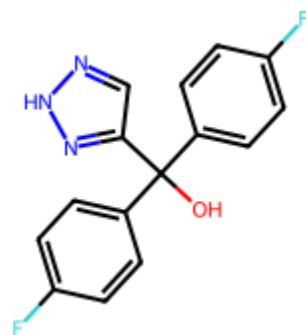
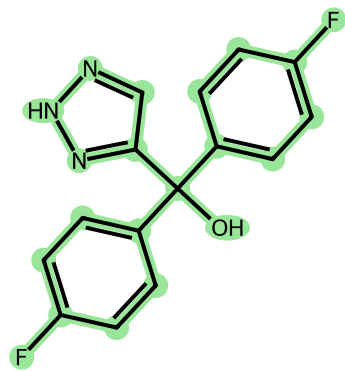
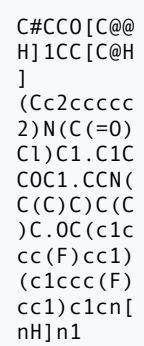
MCS Match Group 1 (4 predictions) — best score -0.0303

MCS core (21 atoms):



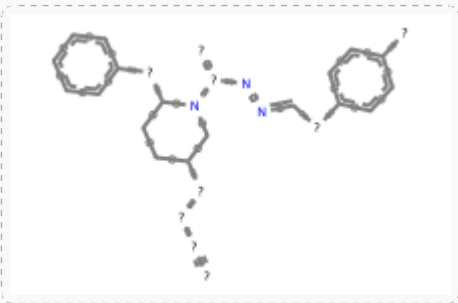
Rank	Score	Reactants SMILES	Largest Fragment (MCS highlighted)	All Reactants
Top-3	-0.0303	<chem>C#CCO[C@@H]1CC[C@H](Cc2ccccc2)N(C(=O)C1)C1.CCN(C(C)C)C(C)C.OC(c1ccc(F)cc1)(c1ccc(F)cc1)c1cn[nH]n1</chem>		
Top-4	-0.0317	<chem>C#CCO[C@@H]1CC[C@H](Cc2ccccc2)N(C(=O)C1)C1.C#CCO[C@@H]1CC[C@H](Cc2ccccc2)N(C(=O)C1)C1.CCN(C(C)C)C(C)C.OC(c1ccc(F)cc1)(c1ccc(F)cc1)c1cn[nH]n1</chem>		
Top-6	-0.0331	<chem>C#CCO[C@@H]1CC[C@H](Cc2ccccc2)N(C(=O)C1)C1.OC(c1ccc(F)cc1)(c1ccc(F)cc1)c1cn[nH]n1</chem>		

Top-8 -0.0346



MCS Match Group 2 (6 predictions) — best score -0.0266

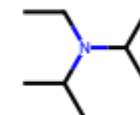
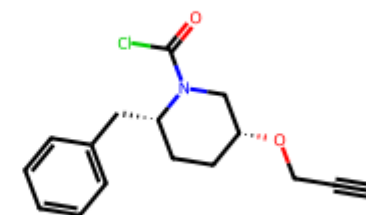
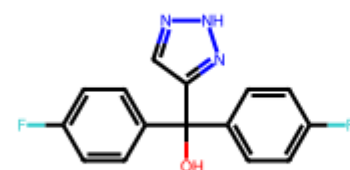
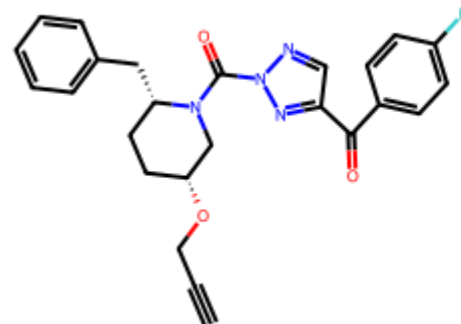
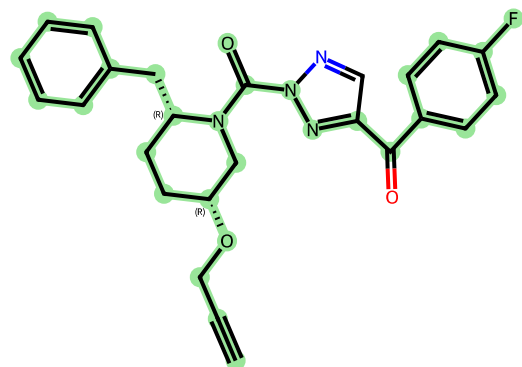
MCS core (30 atoms):



Rank	Score	Reactants SMILES	Largest Fragment (MCS highlighted)	All Reactants		
Top-1	-0.0266	<chem>C#CCO[C@@H]1CC[C@H](C2=CC=CC=C2)N(C(=O)C1)C1.C#CCO[C@@H]1CC[C@H](C2=CC=CC=C2)N(C(=O)C1)C1</chem>				
Top-2	-0.0292	<chem>C#CCO[C@@H]1CC[C@H](C2=CC=CC=C2)N(C(=O)C1)C1.C#CCO[C@@H]1CC[C@H](C2=CC=CC=C2)N(C(=O)C1)C1</chem>				
Top-5	-0.0319	<chem>C#CCO[C@@H]1CC[C@H](C2=CC=CC=C2)N(C(=O)C1)C1.C#CCO[C@@H]1CC[C@H](C2=CC=CC=C2)N(C(=O)C1)C1</chem>				

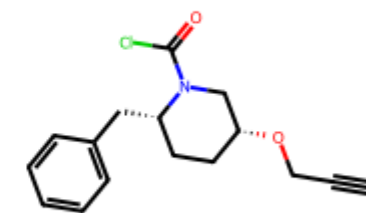
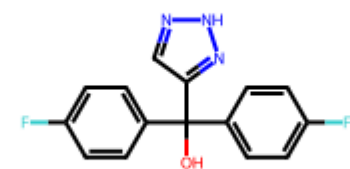
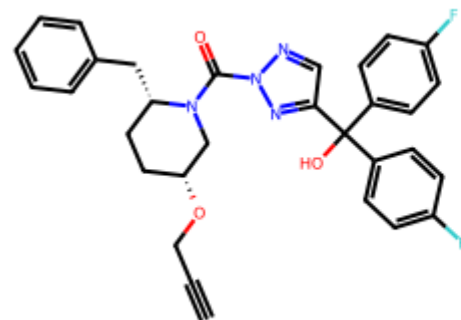
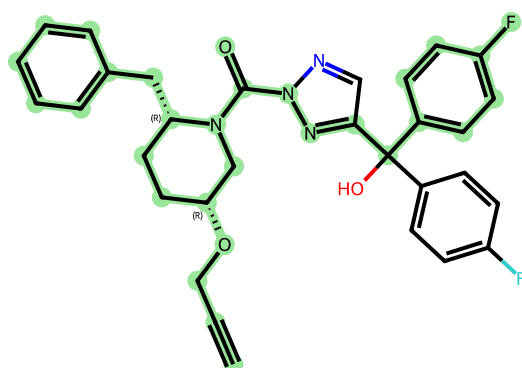
Top-7 -0.0338

```
C#CCO[C@@H]1CC[C@H](C2=CC=CC=C2)N(C(=O)C1)C#CCO[C@H]1CC[C@H](C2=CC=CC=C2)N(C(=O)n2ncc(C(=O)c3ccc(F)cc3)n2)C1.CCN(C(C)C)C(C)C.COC(c1ccc(F)cc1)(c1ccc(F)cc1)c1cn[nH]n1
```



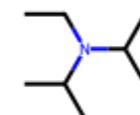
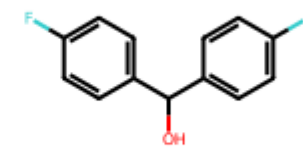
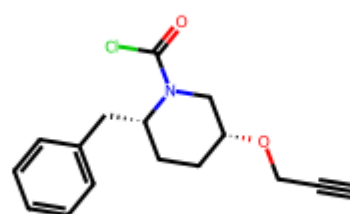
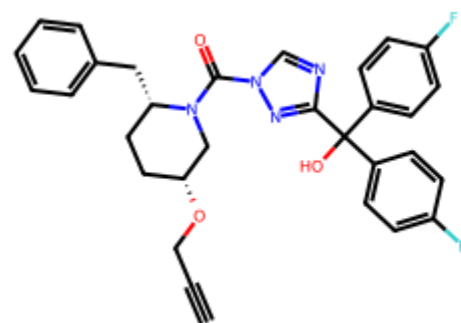
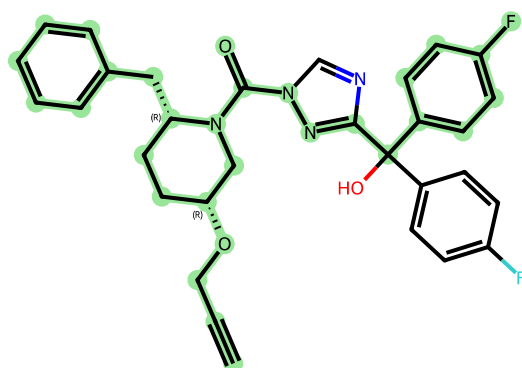
Top-9 -0.0358

```
C#CCO[C@@H]1CC[C@H](C2=CC=CC=C2)N(C(=O)C1)C#CCO[C@H]1CC[C@H](C2=CC=CC=C2)N(C(=O)n2ncc(C(=O)c3ccc(F)cc3)c3ccc(F)cc3)n2)C1.CCN(C(C)C)C(C)C.COC(c1ccc(F)cc1)(c1ccc(F)cc1)c1cn[nH]n1
```



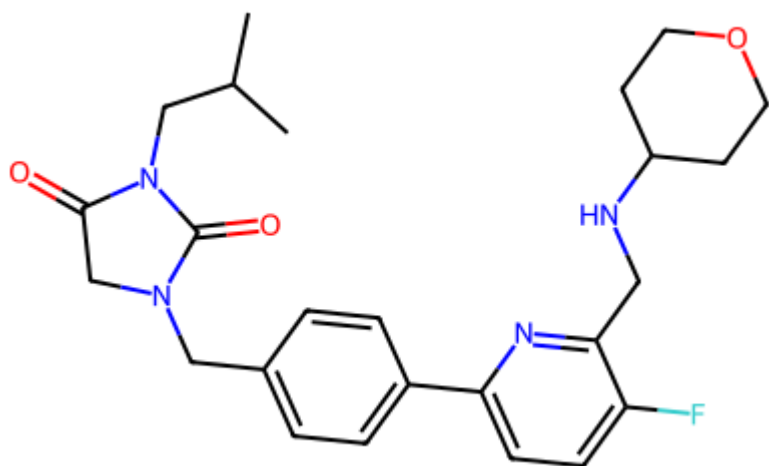
Top-10 -0.0400

```
C#CCO[C@@H]1CC[C@H](C2=CC=CC=C2)N(C(=O)C1)C#CCO[C@H]1CC[C@H](C2=CC=CC=C2)N(C(=O)n2cnc(C(=O)c3ccc(F)cc3)c3ccc(F)cc3)n2)C1.CCN(C(C)C)C(C)C.COC(c1ccc(F)cc1)(c1ccc(F)cc1)c1cn[nH]n1
```



Product 2

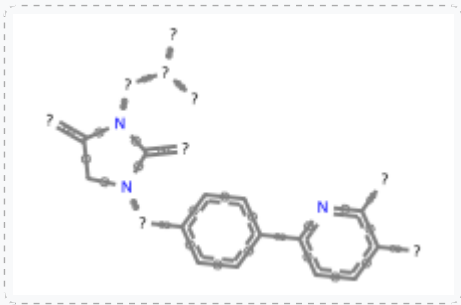
CC(C)CN1C(=O)CN(Cc2ccc(-c3ccc(F)c(CNC4CCOCC4)n3)cc2)C1=O



10 predictions → 1 MCS match group

MCS Match Group 1 (10 predictions) — best score -0.0342

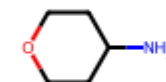
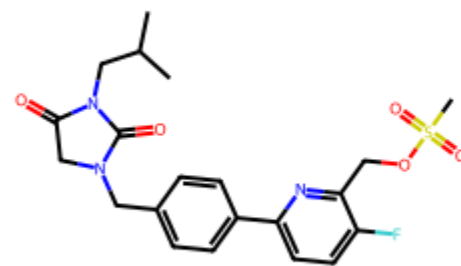
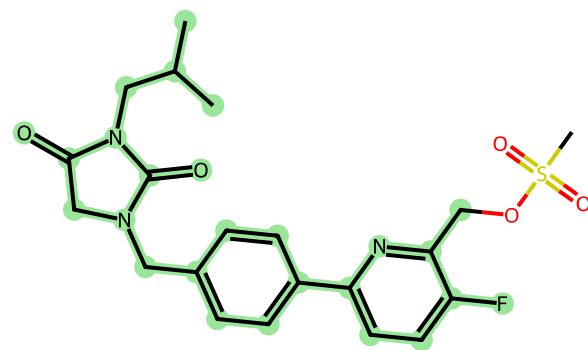
MCS core (26 atoms):



Rank	Score	Reactants SMILES	Largest Fragment (MCS highlighted)	All Reactants
Top-1	-0.0342	<chem>CC(C)CN1C(=O)CN(Cc2ccc(-c3ccc(F)c(CN(C(=O)OC(C)(C)C4CCOCC4)n3)c2)C1=O.C1CC1.O=C(O)C(F)(F)F</chem>		
Top-2	-0.0394	<chem>CC(C)CN1C(=O)CN(Cc2ccc(-c3ccc(F)c(CO)n3)cc2)C1=O.CCNC(C(C)C)C(C)C.CS(=O)(=O)C1.NC1CCOCC1.O=C([O-])[O-].[K+].[K+]</chem>		
Top-3	-0.0403	<chem>CC(C)CN1C(=O)CN(Cc2ccc(-c3ccc(F)c(CO)n3)cc2)C1=O.CCNC(C(C)C)C(C)C.NC1C(=O)CC1.O=S(=O)(OS(=O)(=O)C(F)(F)F)C(F)(F)F</chem>		

Top-4 -0.0408

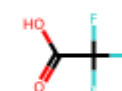
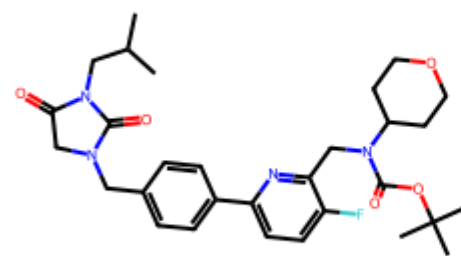
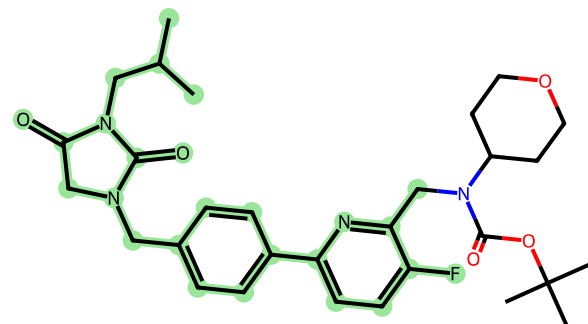
CC(C)CN1C(=O)CN(Cc2ccc(-c3ccc(F)c(COS(C)(=O)=O)n3)cc2)C1=O.NC1CCOCC1.O=C([O-])[O-].[K+].[K+]



K⁺

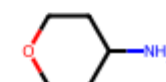
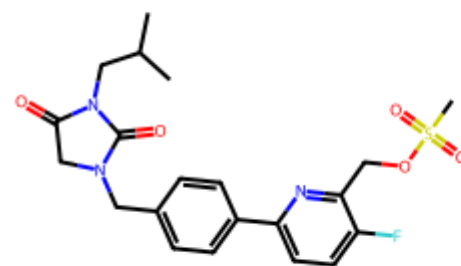
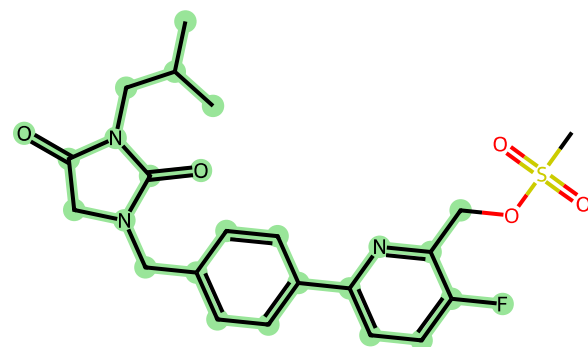
Top-5 -0.0418

CC(C)CN1C(=O)CN(Cc2ccc(-c3ccc(F)c(CN(C(=O)OC(C)(C)C4CCOCC4)n3)c2)C1=O.O=C(O)C(F)(F)F



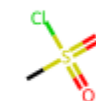
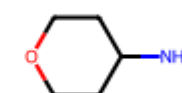
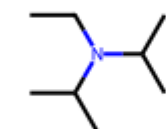
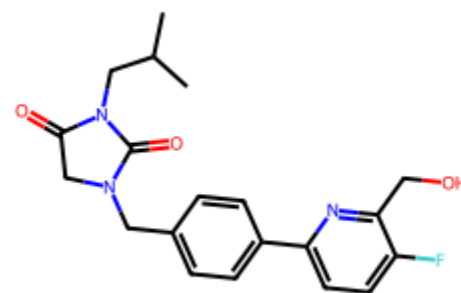
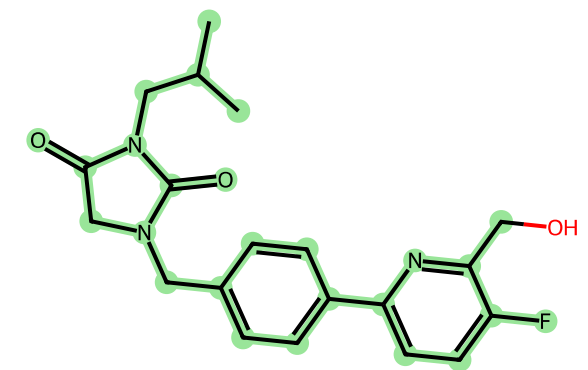
Top-6 -0.0446

CC(C)CN1C(=O)CN(Cc2ccc(-c3ccc(F)c(COS(C)(=O)=O)n3)cc2)C1=O.NC1CCOCC1



Top-7 -0.0486

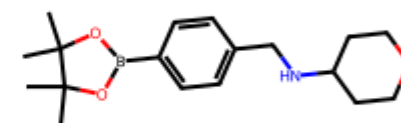
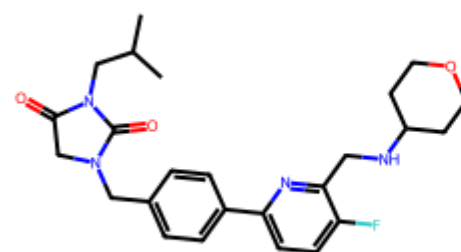
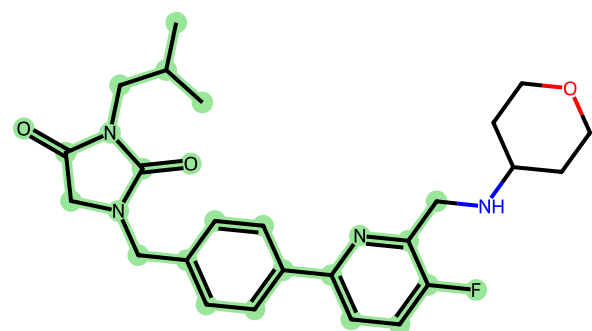
CC(C)CN1C(=O)CN(Cc2ccc(-c3ccc(F)c(CO)n3)cc2)C1=O.CCNC(C(C)C)C(C)C.CS(=O)(=O)Cl.NC1CCOCC1



Top-8

-0.0502

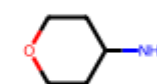
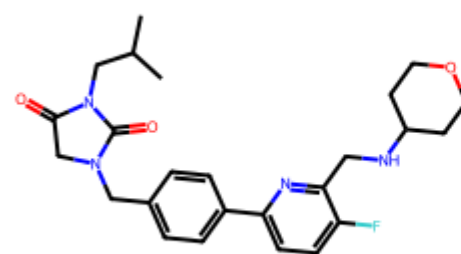
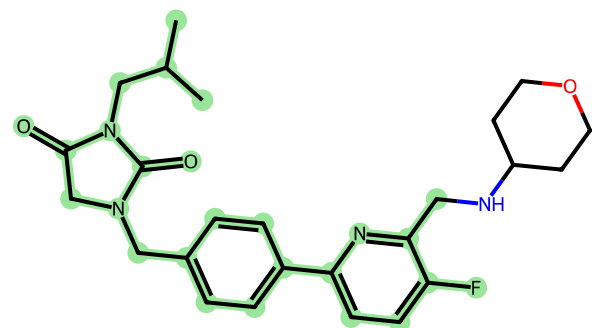
```
CC(C)CN1C
(=O)CN(Cc
2ccc(-
c3ccc(F)c
(CNC4CCOC
C4)n3)cc2
)C1=O.CC1
(C)OB(c2c
cc(CNC3CC
OCC3)cc2)
OC1(C)C
```



Top-9

-0.0545

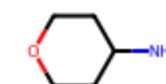
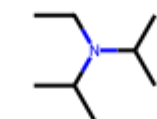
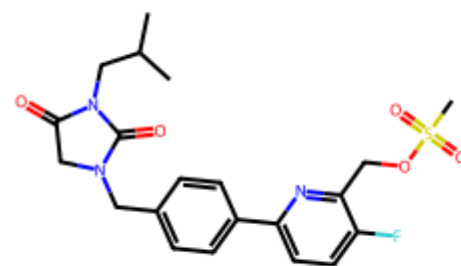
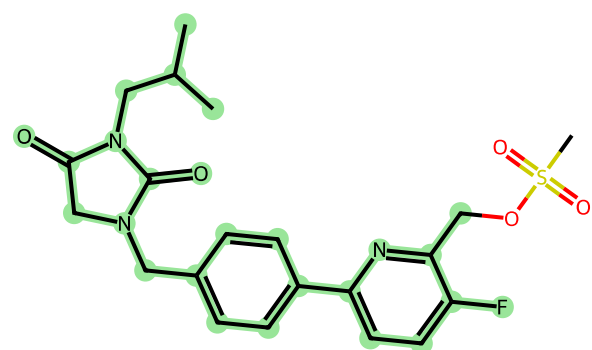
```
CC(C)CN1C
(=O)CN(Cc
2ccc(-
c3ccc(F)c
(CNC4CCOC
C4)n3)cc2
)C1=O.NC1
CCOCC1
```



Top-10

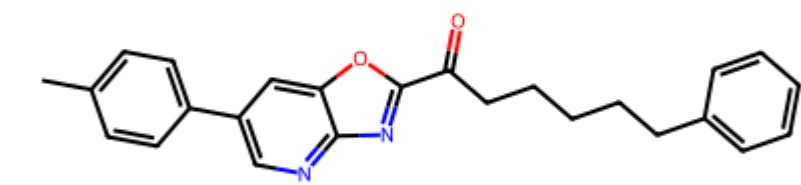
-0.0590

```
CC(C)CN1C
(=O)CN(Cc
2ccc(-
c3ccc(F)c
(COS(C)
(=O)=O)n3
)cc2)C1=O
.CCN(C(C)
C)C(C)C.N
C1CCOCC1
```



Product 3

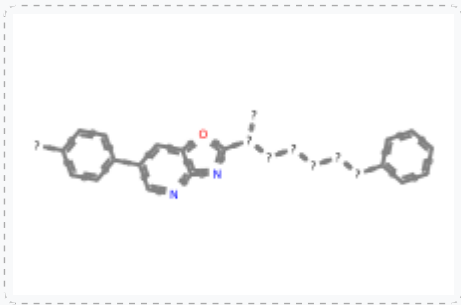
Cc1ccc(-c2cnc3nc(C(=O)CCCCCc4ccccc4)oc3c2)cc1



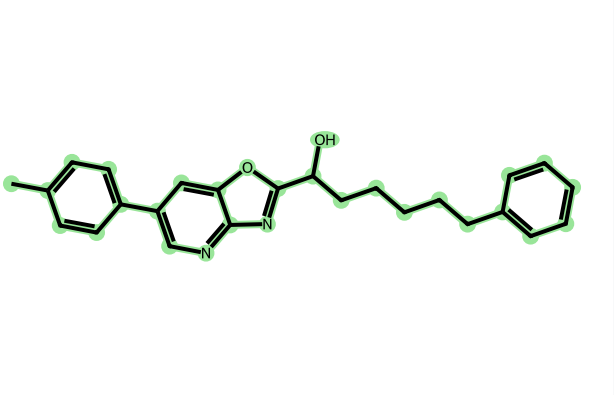
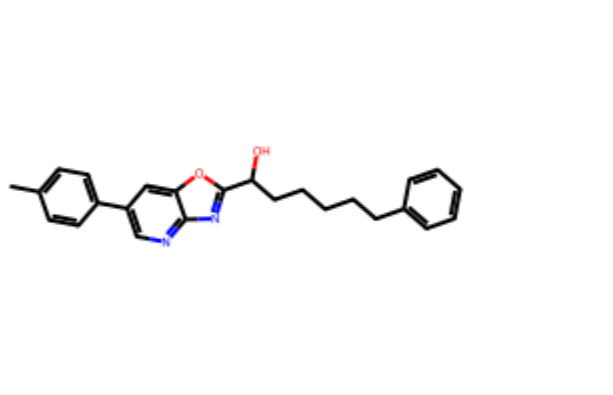
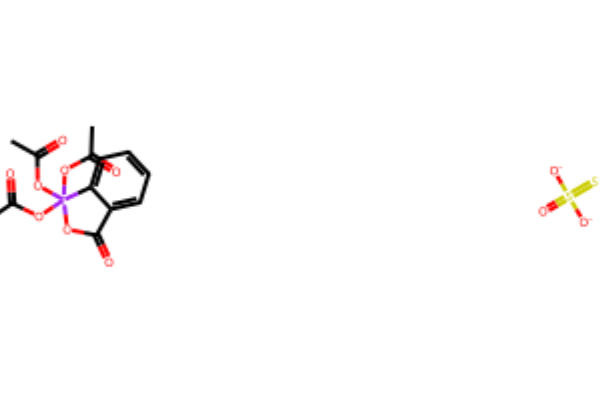
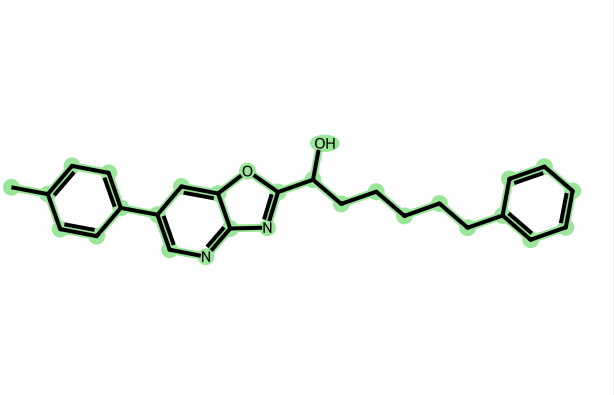
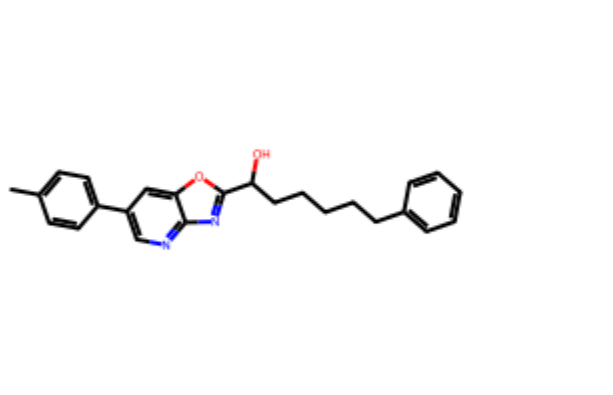
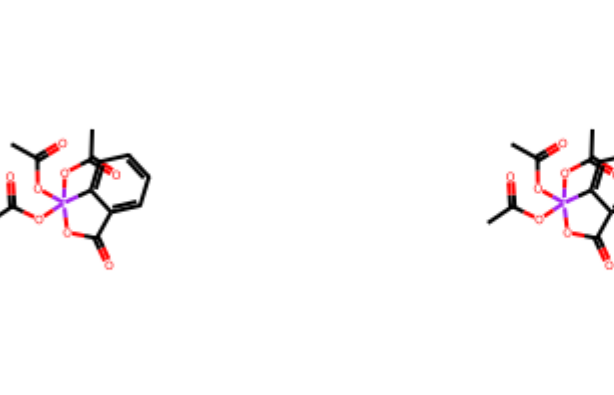
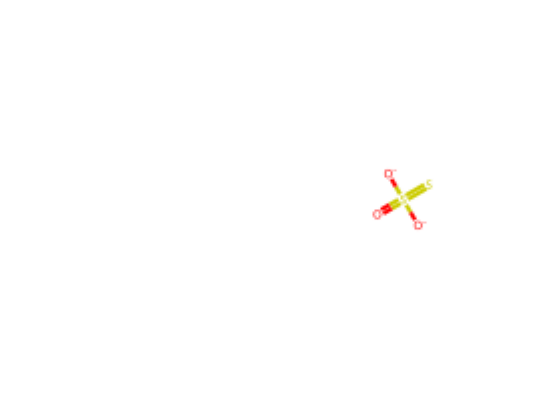
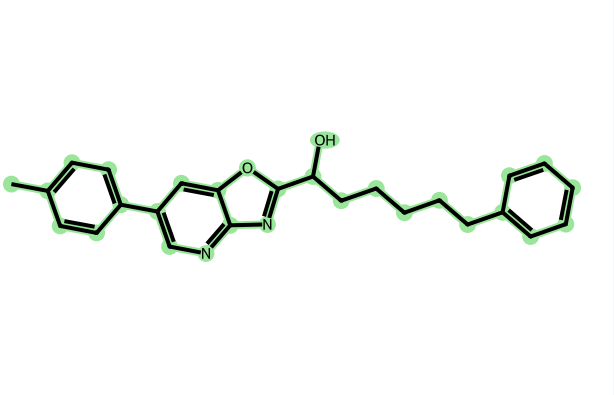
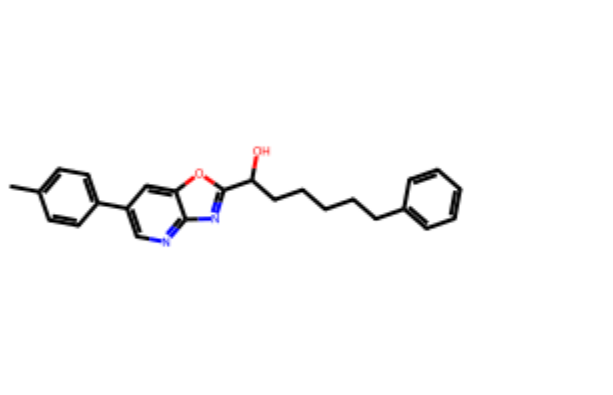
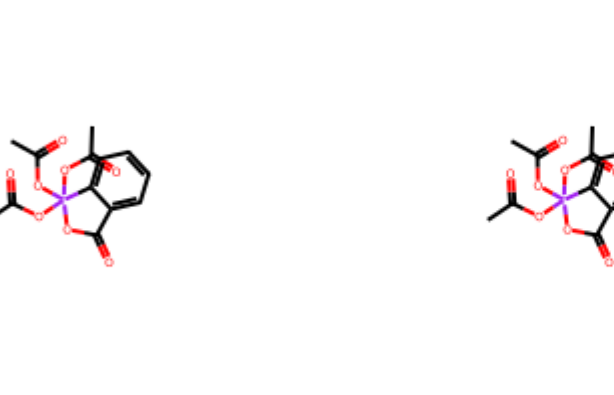
10 predictions → 3 MCS match groups

MCS Match Group 1 (6 predictions) — best score -0.0187

MCS core (29 atoms):

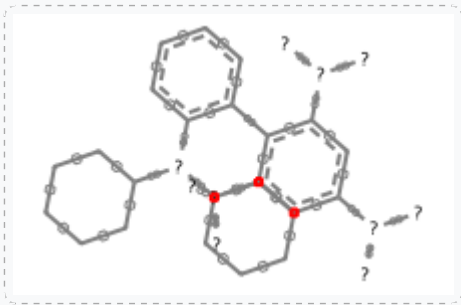


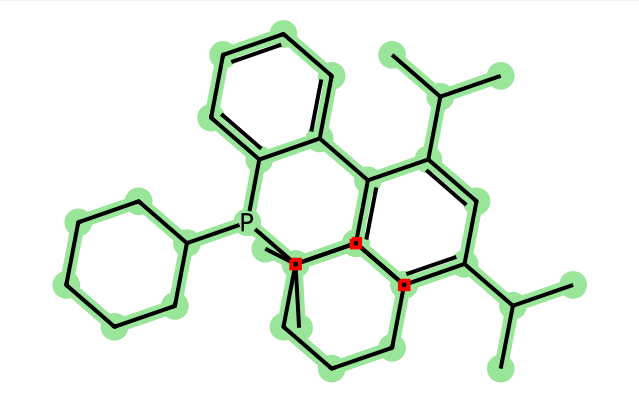
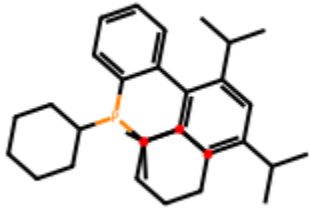
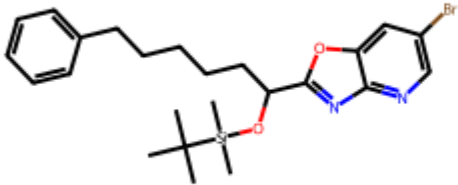
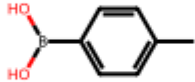
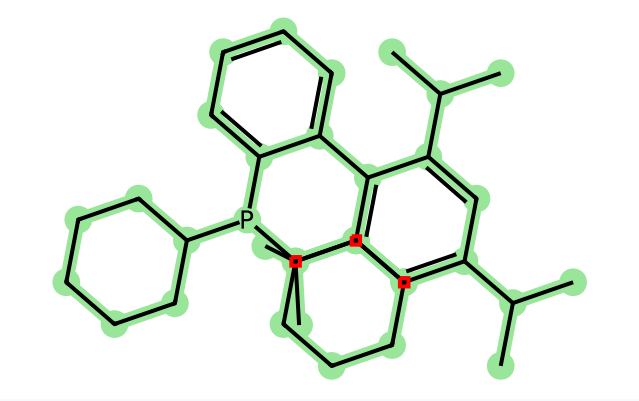
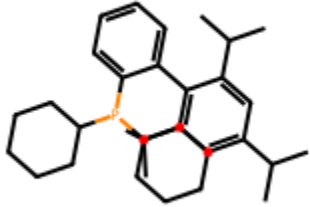
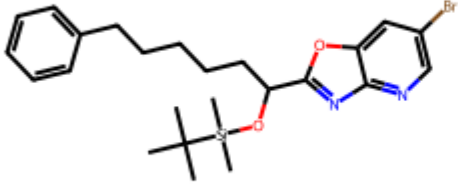
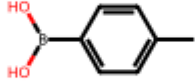
Rank	Score	Reactants SMILES	Largest Fragment (MCS highlighted)	All Reactants			
Top-1	-0.0187	<chem>CC(=O)OI1(OC(C)=O)(OC(C)=O)OC(=O)c2c cccc21.CC(=O)OI1(OC(C)=O)(OC(C)=O)OC(=O)c2c cccc21.Cc1ccc(-c2cnc3nc(C(0)CCCCc4cccc4)oc3c2)cc1</chem>					
Top-2	-0.0192	<chem>CC(=O)OI1(OC(C)=O)(OC(C)=O)OC(=O)c2c cccc21.Cc1ccc(-c2cnc3nc(C(0)CCCCc4cccc4)oc3c2)cc1</chem>					
Top-4	-0.0329	<chem>CC(=O)OI1(OC(C)=O)(OC(C)=O)OC(=O)c2c cccc21.CC(=O)OI1(OC(C)=O)(OC(C)=O)OC(=O)c2c cccc21.Cc1ccc(-c2cnc3nc(C(0)CCCCc4cccc4)oc3c2)cc1.Cc1ccc(-c2cnc3nc(C(0)CCCCc4cccc4)oc3c2)cc1</chem>					

Top-5	-0.0332	<chem>CC(=O)OI1(OC(C)=O)(OC(C)=O)OC(=O)c2ccccc21.Cc1ccc(-c2cnc3nc(C(=O)CCCCc4ccccc4)oc3c2)cc1.O=C([O-])O.O=S([O-])=S.[Na+].[Na+].[Na+]</chem>    
Top-8	-0.0392	<chem>CC(=O)OI1(OC(C)=O)(OC(C)=O)OC(=O)c2ccccc21.CC(=O)OI1(OC(C)=O)(OC(C)=O)OC(=O)c2ccccc21.Cc1ccc(-c2cnc3nc(C(=O)CCCCc4ccccc4)oc3c2)cc1.O=C([O-])O.O=S([O-])=S.[Na+]</chem>    
Top-10	-0.0438	<chem>CC(=O)OI1(OC(C)=O)(OC(C)=O)OC(=O)c2ccccc21.CC(=O)OI1(OC(C)=O)(OC(C)=O)OC(=O)c2ccccc21.Cc1ccc(-c2cnc3nc(C(=O)CCCCc4ccccc4)oc3c2)cc1.O=S([O-])=S.[Na+].[Na+]</chem>    

MCS Match Group 2 (3 predictions) — best score -0.0365

MCS core (34 atoms):



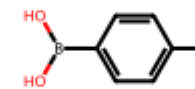
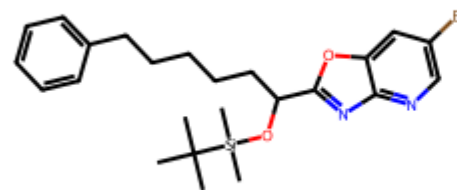
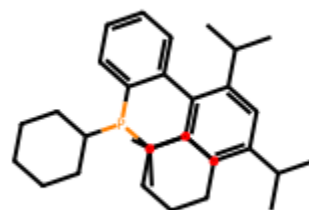
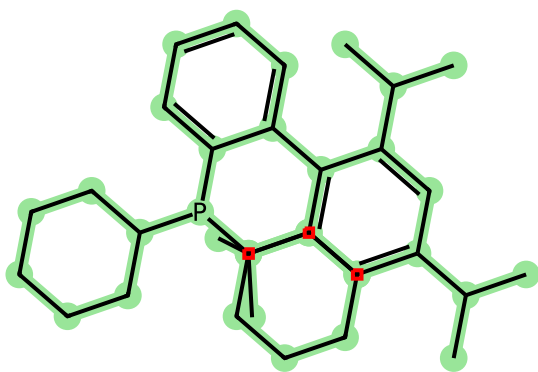
Rank	Score	Reactants SMILES	Largest Fragment (MCS highlighted)	All Reactants
Top-6	-0.0365	<chem>CC(C)(C)[Si](C)(C)OC(CCC(Cc1ccccc1)c1nc2nc(c(Br)cc2o1)CC(C)c1cc(C(C)C)c(-c2cccc2P(C2CCCC2)C2CCCC2)c(C(C)C)c1.Cc1ccc(B(O)O)cc1.O=P([O-])([O-])([O-])[K+].[K+]</chem>		   
Top-7	-0.0380	<chem>CC(C)(C)[Si](C)(C)OC(CCC(Cc1ccccc1)c1nc2nc(c(Br)cc2o1)CC(C)c1cc(C(C)C)c(-c2cccc2P(C2CCCC2)C2CCCC2)c(C(C)C)c1.Cc1ccc(B(O)O)cc1.O=C([O-])[O-].[K+].[K+]</chem>		   

Top-9 -0.0412

```

CC(C)(C)
[Si](C)
(C)OC(CCC
CCc1cccc
1)c1nc2nc
c(Br)cc2o
1.CC(C)c1
cc(C(C)C)
c(-
c2cccc2P
(C2CCCCC2
)C2CCCCC2
)c(C(C)C)
c1.Cc1ccc
(B(0)O)cc
1.O=C([O-
]) [O-] .
[Cs+] .
[Cs+]

```

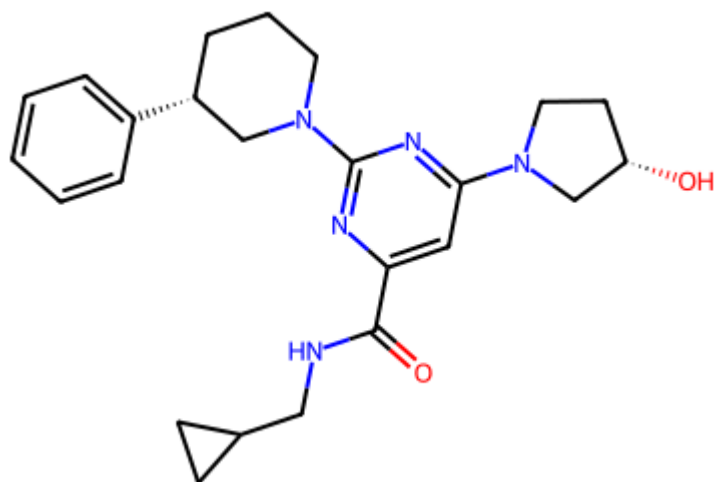


MCS Match Group 3 (1 prediction) — best score -0.0316

Rank	Score	Reactants SMILES	Largest Fragment (MCS highlighted)	All Reactants
Top-3	-0.0316	<pre> CC(=O)O11 (OC(C)=O) (OC(C)=O) OC(=O)c2c cccc21.CC (=O)O11(O C(C)=O) (OC(C)=O) OC(=O)c2c cccc21.CC (=O)O11(O C(C)=O) (OC(C)=O) OC(=O)c2c cccc21.Cc 1ccc(- c2cnc3nc(C(0)CCCC c4cccc4) oc3c2)c </pre>		

Product 4

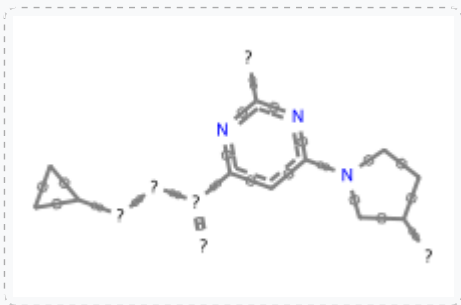
O=C(NCC1CC1)c1cc(N2CC[C@H](O)C2)nc(N2CCC[C@H](c3ccccc3)C2)n1



10 predictions → 1 MCS match group

MCS Match Group 1 (10 predictions) — best score -0.0335

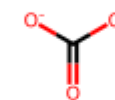
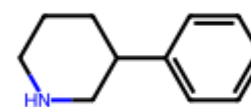
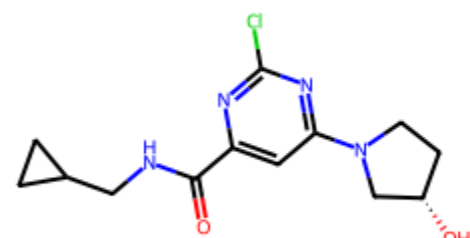
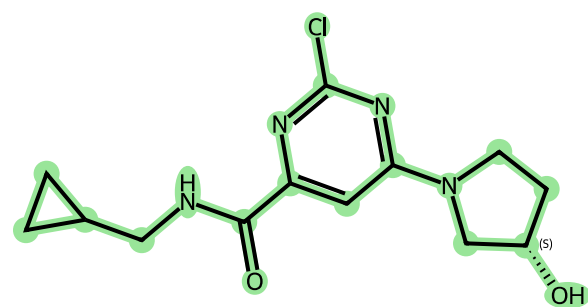
MCS core (20 atoms):



Rank	Score	Reactants SMILES	Largest Fragment (MCS highlighted)	All Reactants
Top-1	-0.0335	<chem>CCN(C(C)C)C(C)C.Cl.O=C(NCC1CC1)c1cc(N2CC[C@H](O)C2)nc(Cl)n1.c1ccc([C@H]2CCNC2)c1</chem>		
Top-2	-0.0392	<chem>Cl.O=C(NC1CC1)c1cc(N2CC[C@H](O)C2)nc(Cl)n1.O=C([O-])[O-].[K+].[K+].c1ccc(C2CCNC2)cc1</chem>		
Top-3	-0.0404	<chem>O=C(NCC1C1)c1cc(N2CC[C@H](O)C2)nc(Cl)n1.O=C([O-])[O-].[Cs+].[Cs+].c1ccc(C2CCCN2)cc1</chem>		

Top-4 -0.0405

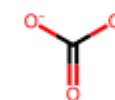
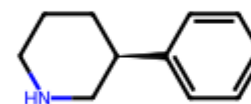
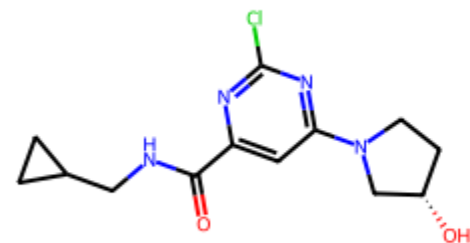
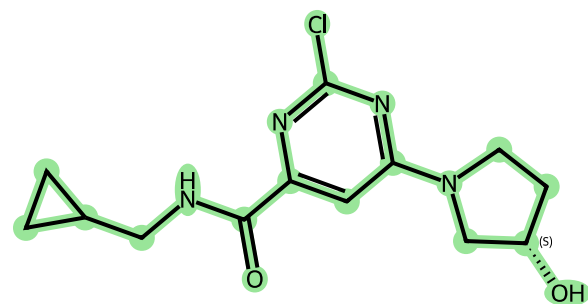
```
C1.O=C(NC
C1CC1)c1c
c(N2CC[C@
H]
(O)C2)nc(
C1)n1.O=C
([O-])
[O-].
[Cs+].
[Cs+].c1c
cc(C2CCCN
C2)cc1
```



HCl

Top-5 -0.0438

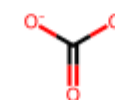
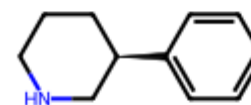
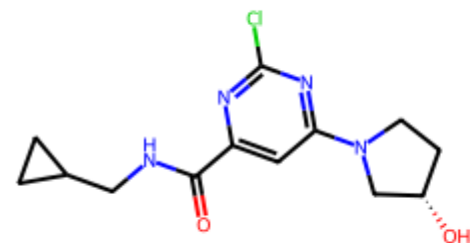
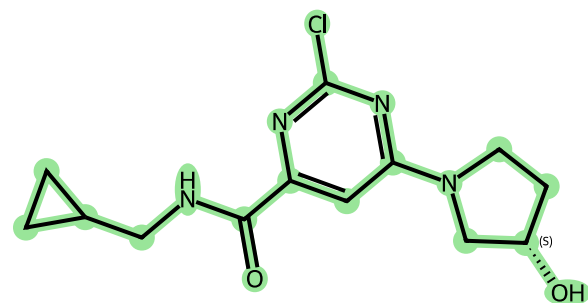
```
C1.O=C(NC
C1CC1)c1c
c(N2CC[C@
H]
(O)C2)nc(
C1)n1.O=C
([O-])
[O-].
[Cs+].
[Cs+].c1c
cc([C@@H]
2CCCN2)c
c1
```



HCl

Top-6 -0.0450

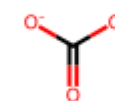
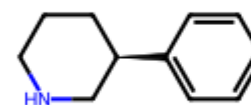
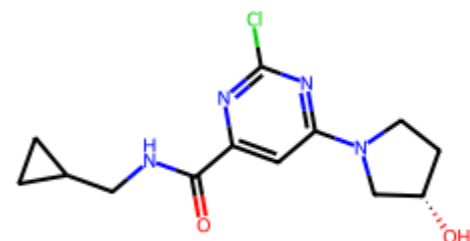
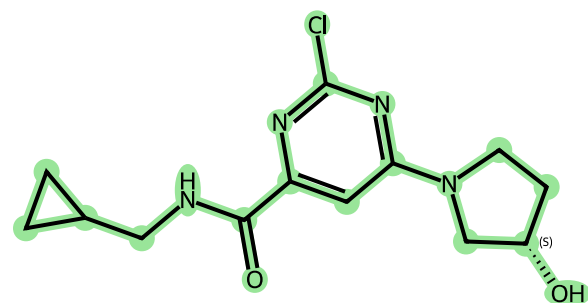
```
O=C(NCC1C
C1)c1cc(N
2CC[C@H]
(O)C2)nc(
C1)n1.O=C
([O-])
[O-].
[Cs+].
[Cs+].c1c
cc([C@@H]
2CCCN2)c
c1
```



Cs⁺

Top-7 -0.0452

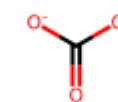
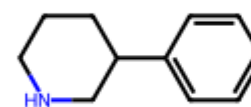
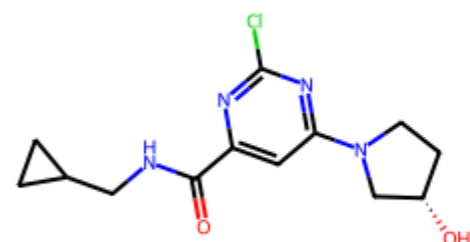
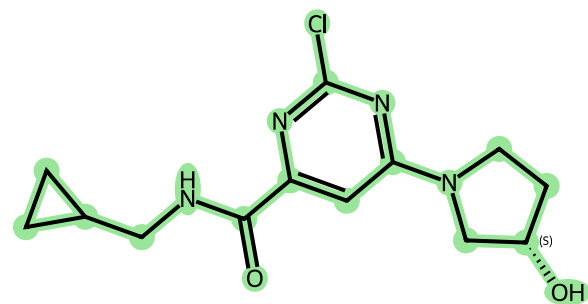
```
C1.O=C(NC
C1CC1)c1c
c(N2CC[C@
H]
(O)C2)nc(
C1)n1.O=C
([O-])
[O-].
[K+].
[K+].c1cc
c([C@@H]2
CCCN2)cc
1
```



HCl

Top-8 -0.0468

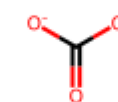
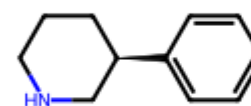
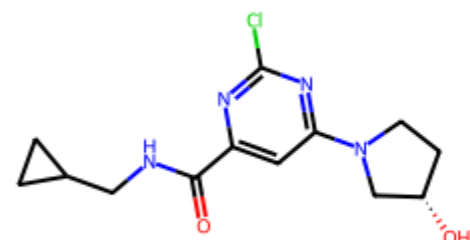
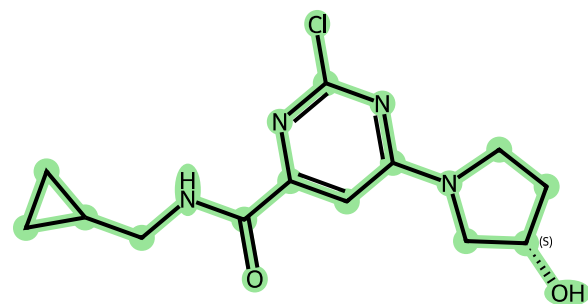
```
O=C(NCC1C
C1)c1cc(N
2CC[C@H]
(O)C2)nc(
C1)n1.O=C
([O-])
[O-].
[K+].
[K+].c1cc
c(C2CCNC
2)cc1
```



K⁺

Top-9 -0.0497

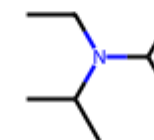
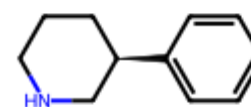
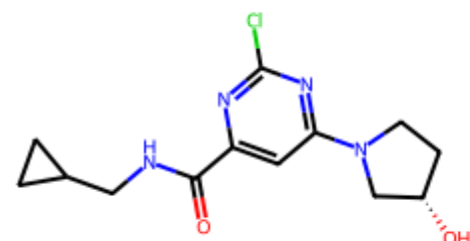
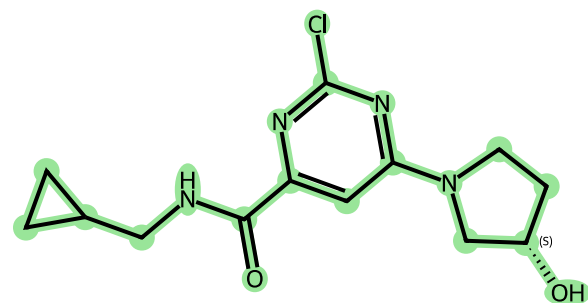
```
O=C(NCC1C
C1)c1cc(N
2CC[C@H]
(O)C2)nc(
C1)n1.O=C
([O-])
[O-].
[K+].
[K+].c1cc
c([C@H]2
CCNC2)cc
1
```



K⁺

Top-10 -0.0515

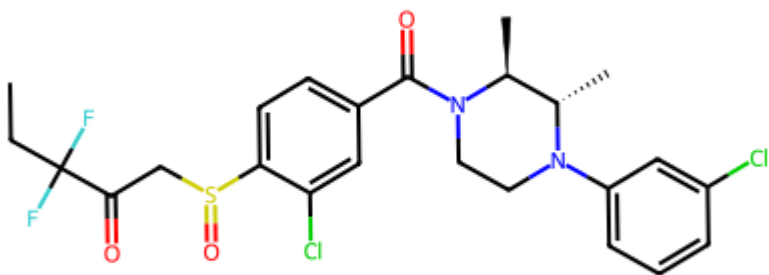
```
CCN(C(C)C
)C(C)C.Cl
.O.O=C(NC
C1CC1)c1c
c(N2CC[C@
H]
(O)C2)nc(
C1)n1.c1c
cc([C@H]
2CCNC2)c
c1
```



HCl

Product 5

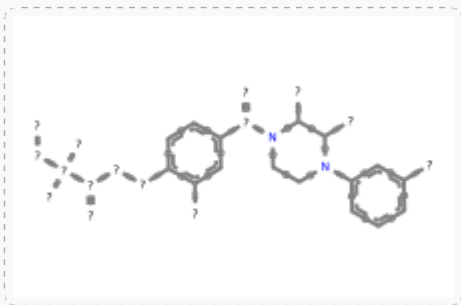
CCC(F)(F)C(=O)CS(=O)c1ccc(C(=O)N2CCN(c3cccc(Cl)c3)[C@@H](C)[C@@H]2C)cc1Cl



10 predictions → 1 MCS match group

MCS Match Group 1 (10 predictions) — best score -0.0252

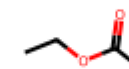
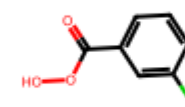
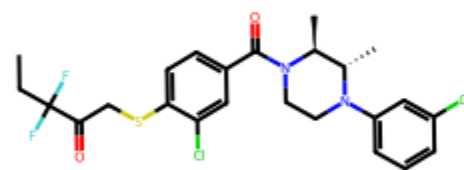
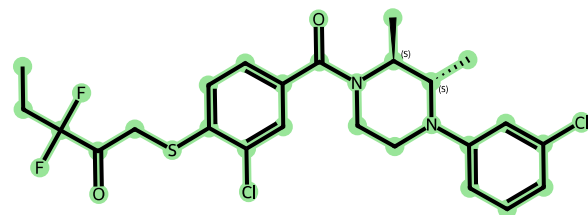
MCS core (33 atoms):



Rank	Score	Reactants SMILES	Largest Fragment (MCS highlighted)	All Reactants
Top-1	-0.0252	<chem>CCC(F)(F)C(=O)CSc1ccc(C(=O)N2CCN(c3cccc(Cl)c3)[C@@H](C)[C@@H]2C)cc1Cl.CO.[Na+].[O-].[I+3].[O-].[O-].[O-].[O-]</chem>		
Top-2	-0.0286	<chem>CCC(F)(F)C(=O)CSc1ccc(C(=O)N2CCN(c3cccc(Cl)c3)[C@@H](C)[C@@H]2C)cc1Cl.O=C(O)c1ccc(Cl)c1.O.[Na+]</chem>		
Top-3	-0.0314	<chem>CCC(F)(F)C(=O)CSc1ccc(C(=O)N2CCN(c3cccc(Cl)c3)[C@@H](C)[C@@H]2C)cc1Cl.CCC(F)(F)C(=O)CSc1ccc(C(=O)N2CCN(c3cccc(Cl)c3)[C@@H](C)[C@@H]2C)cc1Cl.O=C(O)c1ccc(Cl)c1</chem>		

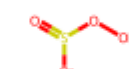
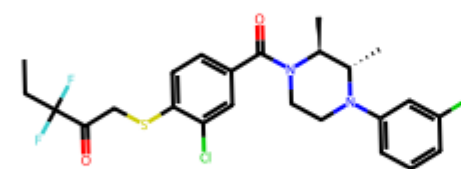
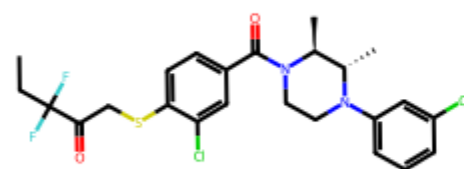
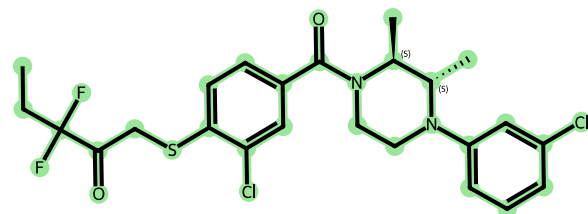
Top-4 -0.0329

```
CCC(F)
(F)C(=O)C
Sc1ccc(C(
=O)N2CCN(
c3cccc(Cl
)c3)
[C@@H](C)
[C@@H]2C)
cc1Cl.CC0
C(C)=O.O=
C(OO)c1cc
cc(Cl)c1.
O=C([O-])
O.[Na+]
```



Top-5 -0.0335

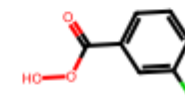
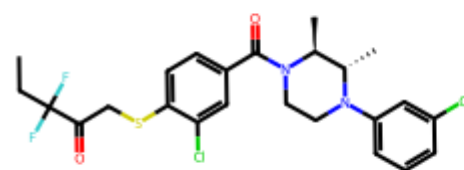
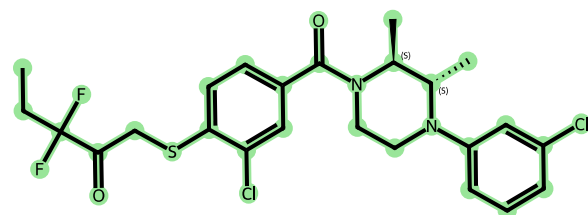
```
CCC(F)
(F)C(=O)C
Sc1ccc(C(
=O)N2CCN(
c3cccc(Cl
)c3)
[C@@H](C)
[C@@H]2C)
cc1Cl.CCC
(F)
(F)C(=O)C
Sc1ccc(C(
=O)N2CCN(
c3cccc(Cl
)c3)
[C@@H](C)
[C@@H]2C)
cc1Cl.O=S
([O-])OO.
[K+]
```



K⁺

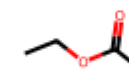
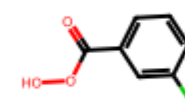
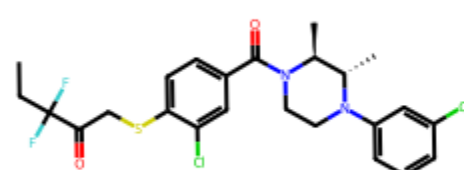
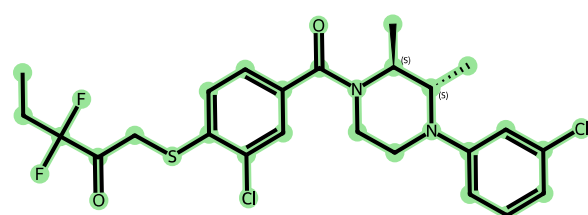
Top-6 -0.0366

```
CCC(F)
(F)C(=O)C
Sc1ccc(C(
=O)N2CCN(
c3cccc(Cl
)c3)
[C@@H](C)
[C@@H]2C)
cc1Cl.O=C
(OO)c1ccc
c(Cl)c1
```



Top-7 -0.0376

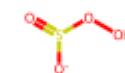
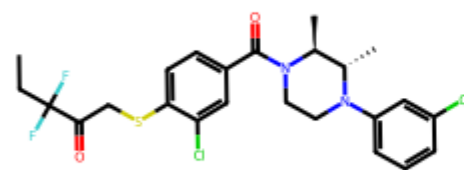
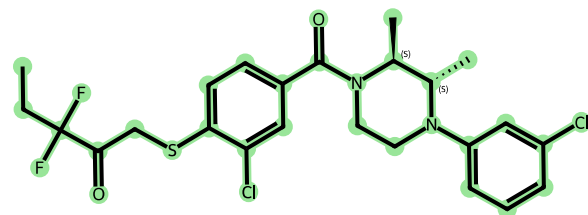
```
CCC(F)
(F)C(=O)C
Sc1ccc(C(
=O)N2CCN(
c3cccc(Cl
)c3)
[C@@H](C)
[C@@H]2C)
cc1Cl.CC0
C(C)=O.O.
O=C(OO)c1
cccc(Cl)c
1.O=C([O-
])O.[Na+]
```



Top-8

-0.0383

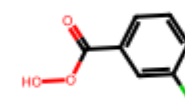
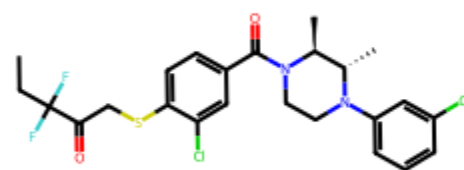
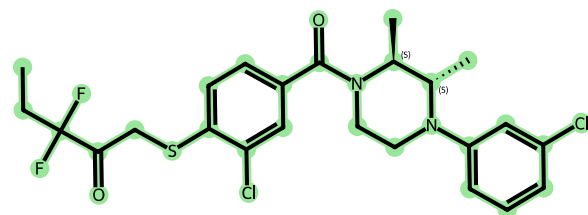
```
CCC(F)
(F)C(=O)C
Sc1ccc(C(
=O)N2CCN(
c3cccc(Cl
)c3)
[C@@H](C)
[C@@H]2C)
cc1Cl.O=S
([O-])O0.
[K+]
```

K⁺

Top-9

-0.0384

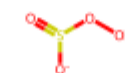
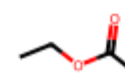
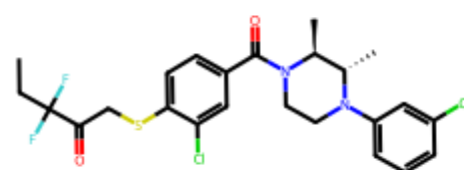
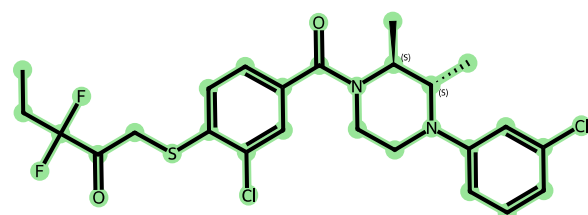
```
CCC(F)
(F)C(=O)C
Sc1ccc(C(
=O)N2CCN(
c3cccc(Cl
)c3)
[C@@H](C)
[C@@H]2C)
cc1Cl.O=C
(OO)c1ccc
c(Cl)c1.O
=S([O-])
([O-])=S.
[Na+].
[Na+]
```

Na⁺

Top-10

-0.0395

```
CCC(F)
(F)C(=O)C
Sc1ccc(C(
=O)N2CCN(
c3cccc(Cl
)c3)
[C@@H](C)
[C@@H]2C)
cc1Cl.CC0
C(C)=O.O.
O=S([O-])
O0.[K+]
```

H₂O

Summary

Product	# Predictions	# MCS Match Groups	Best Score
C#CCO[C@@H]1CC[C@H] (Cc2ccccc2)N(C(=O)n2ncc(C(=O)(c3ccc(F)cc3)...	10	2	0.0266
CC(C)CN1C(=O)CN(Cc2ccc(- c3ccc(F)c(CNC4CCOCC4)n3)cc2)C1=O	10	1	0.0342
Cc1ccc(-c2cnc3nc(C(=O)CCCCc4ccccc4)oc3c2)cc1	10	3	0.0187
O=C(NCC1CC1)c1cc(N2CC[C@H] (O)C2)nc(N2CCC[C@H](c3ccccc3)C2)n...	10	1	0.0335
CCC(F) (F)C(=O)CS(=O)c1ccc(C(=O)N2CCN(c3cccc(Cl)c3) [C@@H](C)[...	10	1	0.0252